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MID-TERM EXAM OF TOOLS OF PROGRAMMING
PROF. BONIFACE MBOUZAO
DURATION: 3 HOURS

NAME AND SURNAME

QUESTION 1 4/5

This is an array T of length 6, answers to the reference to this array a:

4	6	7	3	1	2
0	1	2	3	4	5

For K=2

T[3] = 3
T[K] = T(2) = 7
T[2*K – T[4]] = T[2*2 – 1] = T[4-1] = T[3] = 3 1/1
T[T[5]] = T[5] = 2 ⇒ T[T[5]] = T[2] = 7
T[1] = 6

Give the type of error in each case:

CASE	TYPE OF ERROR
m'Boniface. I call B	Syntax error
The variable X is 0, and the result is 3/X R	Logical error ✗ Runtime
This error is more subtle S	Syntax error
All balloons are round. If this object is not a balloon, then it is not round L	Logical error

1.50/2

Suppose that X = 8 and Y = 6

EXPRESSION	VALUE
((X > 0) AND (NOTY = 0))	T
(X > 0) AND ((X < Y) OR (Y = 0))	T ✗ F
(NOT X > 0) OR ((X < Y) AND (Y = 0))	F
NOT((X > 0) OR ((X < Y) AND (Y = 0)))	F

1.5/2

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QUESTION 2: 2.5/4

Write an algorithm that takes 4 positive integers and combines (joins) them into a single number.
Ex : with 11, 35, 200, and 7 the algorithm should produce 11352007.
Draw your algorithm with the above data.
You can assume that you have the algorithm : **C ← Join(A, B)**

Givens: A, B, C, D (4 positive integers)
Result: Join2 (Joins the 4 positive integers)
Intermediates: first (Joins the first 2 positive integers).
second (Joins the second 2 positive integers.
Header:
Body:
Header: Join ← Join the 4 positive integers (A, B, C, D).
Module: First ← ~~A B~~ Join(A, B)
Second ← ~~C D~~ Join(C, D)
~~All Header~~ ← Join ~~Join~~ (First, second).
Return (Join)

You may assume there is available an algorithm:

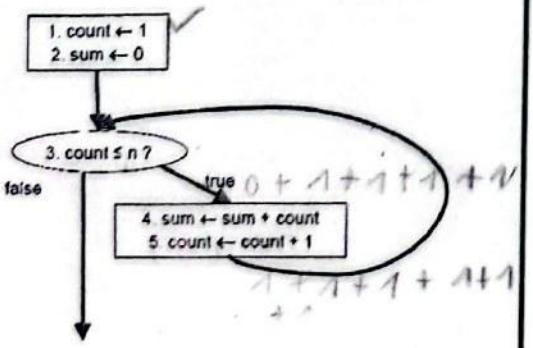
c ← join(a. b)
Givens: a. b. two positive integers
Result: c is the number having the digits in a followed by the digits in b.

w	← 11
x	← 35
y	← 200
z	← 7
wx	1135
wy	2007
allJoined	11352007

QUESTION 3 0/3

Trace this algorithm for n = 5, **sum1ToN(5)**

GIVEN: n (a positive integer)
INTERMEDIATE: count (index going from 1 to n)
RESULT: sum (sum of integers 1 to N)
HEADER: sum ← sum1ToN(n)
BODY:



Steps	Givens n	Intermediate (count)	Result
0	5	Null	Null
1	5	1	null
2	5	1	0
T 3	5	1	0
4	5	1	1
5	5	2	1
T 3	5	1	0
4	5	1	2
5	5	3	2
T 3	5	1	0
4	5	1	3
5	5	4	3
T 3	5	1	0
4	5	1	4
5	5	5	4
T 3	5	1	0
4	5	1	5
5	5	6	5.

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QUESTION 4 4/4

1. Find the maximum between three given numbers X, Y, and Z.
   a. Version 1: with nested tests
   b. Version 2: sequence of tests
2. Trace your sequence tests with: MAX3(6, 9, 12)

1)

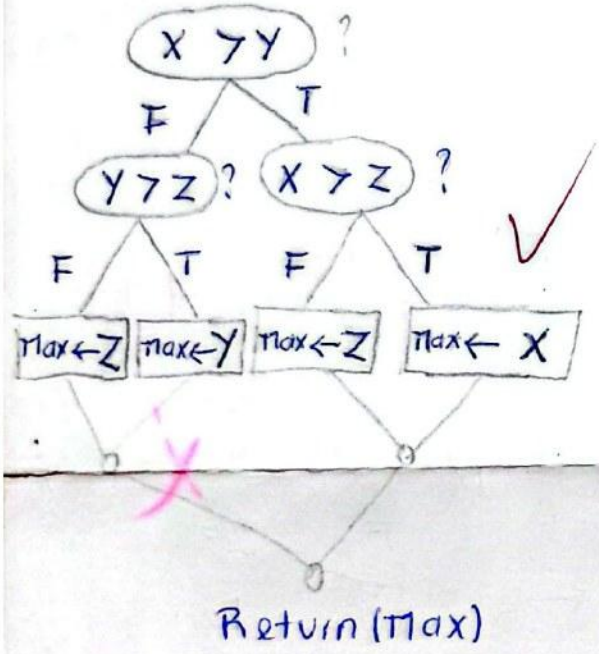
Givens; X, Y, Z (three given numbers).

Results; Max (Maximum between X, Y, Z).

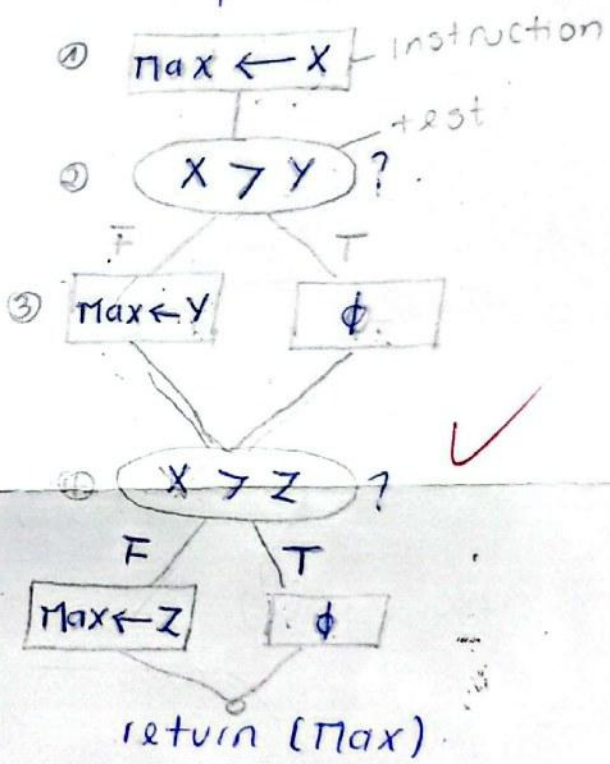
Header: Max ← Maximum between 3 numbers (X, Y, Z).

Module:

Nested Test



Sequence Test.



2)

Steps	X	Y	Z	Givens	~~Interm~~ result (Max)
0	6	9	12		null
1	6	9	12		6
2	6	9	12		6
F 3	6	9	12		9
4	6	9	12		6
F 5	6	9	12		12
					12

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Question 5 3½/4

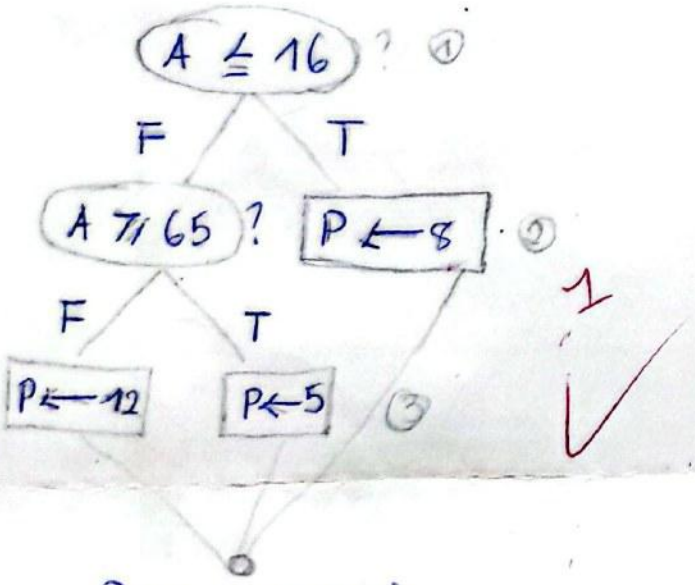
1. Write an algorithm to calculate the price of a ticket for a person when the cost is : $8 for people 16 years old or younger, $5 for people 65 years old or older, and $12 for the others.
   a. Version 1 : nested tests
   b. Version 2 : test sequence
2. Trace: For somebody who is 67 years old.

1) Givens: ~~P (Price of a ticket for a person)~~ ✗
A (Age of person) ✓

Result : Price (Price of ticket for a person depending on his age) | 0.5

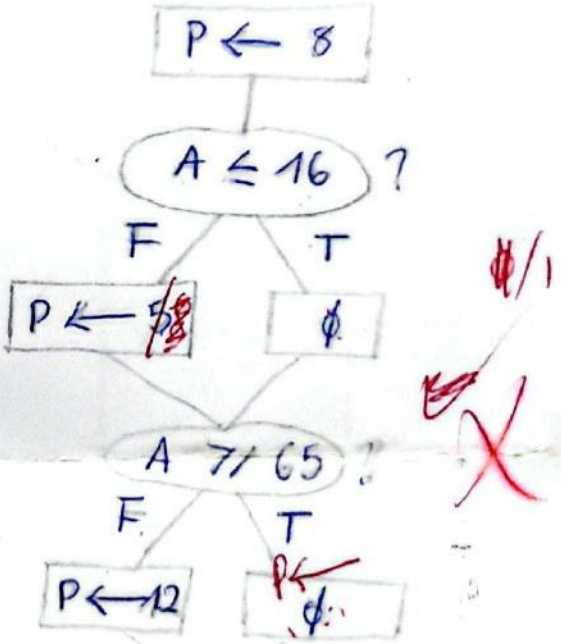
Header : Price ← Price of ticket per person depending on age (A, P) ✗

Module : Nested test



1 ✓

Sequence test



0/1 ✗

2) Tracing for a 67 year old for Nested test

Steps	Givens	result
0	67	NULL
T 1	67	8
T 2	67	8
F 3	67	12
T 3	67	5
		5

1/1 ✗

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